

EX.NO:12 BLOOD DONOR FINDER

program:

```
import java.util.ArrayList;
import java.util.Scanner;

class BloodDonor {
    String name;
    int age;
    String gender;
    String bloodGroup;
    String city;
    String phone;

    // Constructor
    BloodDonor(String name, int age, String gender, String bloodGroup, String
city, String phone) {
        this.name = name;
        this.age = age;
        this.gender = gender;
        this.bloodGroup = bloodGroup;
        this.city = city;
        this.phone = phone;
    }

    // Display donor details
    void display() {
        System.out.println("-----");
        System.out.println("Name      : " + name);
        System.out.println("Age       : " + age);
        System.out.println("Gender    : " + gender);
        System.out.println("Blood Group: " + bloodGroup);
        System.out.println("City      : " + city);
        System.out.println("Phone     : " + phone);
        System.out.println("-----");
    }
}

public class BloodDonorFinder {
```

```

static ArrayList<BloodDonor> donors = new ArrayList<>();

// Add sample donors
static void addSampleDonors() {
    donors.add(new BloodDonor("Arun", 25, "Male", "A+", "Chennai",
"9876543210"));
    donors.add(new BloodDonor("Priya", 22, "Female", "B+", "Cuddalore",
"9876501234"));
    donors.add(new BloodDonor("Karthik", 28, "Male", "O+", "Chennai",
"9876512345"));
    donors.add(new BloodDonor("Divya", 24, "Female", "A+", "Panruti",
"9876523456"));
    donors.add(new BloodDonor("Rahul", 30, "Male", "O-", "Viluppuram",
"9876534567"));
}

// Register a new donor
static void registerDonor(Scanner sc) {
    System.out.print("Enter donor name: ");
    String name = sc.nextLine();

    System.out.print("Enter age: ");
    int age = Integer.parseInt(sc.nextLine());

    System.out.print("Enter gender: ");
    String gender = sc.nextLine();

    System.out.print("Enter blood group: ");
    String bloodGroup = sc.nextLine();

    System.out.print("Enter city: ");
    String city = sc.nextLine();

    System.out.print("Enter phone number: ");
    String phone = sc.nextLine();

    donors.add(new BloodDonor(name, age, gender, bloodGroup, city,
phone));

    System.out.println("\nDonor registered successfully!");
}

```

```

}

// Display all donors
static void displayAllDonors() {
    if (donors.isEmpty()) {
        System.out.println("No donors available.");
        return;
    }
    System.out.println("\n===== ALL BLOOD DONORS =====");
    for (BloodDonor donor : donors) {
        donor.display();
    }
}

// Search by blood group
static void searchByBloodGroup(Scanner sc) {
    System.out.print("Enter required blood group: ");
    String group = sc.nextLine();

    boolean found = false;
    System.out.println("\n===== SEARCH RESULT =====");

    for (BloodDonor donor : donors) {
        if (donor.bloodGroup.equalsIgnoreCase(group)) {
            donor.display();
            found = true;
        }
    }
    if (!found) {
        System.out.println("No donor found for blood group " + group);
    }
}

// Search by city
static void searchByCity(Scanner sc) {
    System.out.print("Enter city: ");
    String city = sc.nextLine();

    boolean found = false;
    System.out.println("\n===== DONORS IN " + city + " =====");
}

```

```

        for (BloodDonor donor : donors) {
            if (donor.city.equalsIgnoreCase(city)) {
                donor.display();
                found = true;
            }
        }
        if (!found) {
            System.out.println("No donors found in " + city);
        }
    }
}

```

```

// Main method
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    addSampleDonors();
    int choice;
    do {

```

```

        System.out.println("\n=====");
        System.out.println("  BLOOD DONOR FINDER");
        System.out.println("=====");
        System.out.println("1. Register Donor");
        System.out.println("2. View All Donors");
        System.out.println("3. Search by Blood Group");
        System.out.println("4. Search by City");
        System.out.println("5. Exit");
        System.out.println("=====");

```

```

        System.out.print("Enter your choice: ");
        choice = Integer.parseInt(sc.nextLine());

```

```

        switch (choice) {
            case 1:
                registerDonor(sc);
                break;
            case 2:
                displayAllDonors();
                break;
            case 3:
                searchByBloodGroup(sc);
                break;

```

```

        case 4:
            searchByCity(sc);
            break;
        case 5:
            System.out.println("Thank you for using Blood Donor Finder!");
            break;
        default:
            System.out.println("Invalid choice!");
    }
} while (choice != 5);
sc.close();
}
}

```

Output:

BLOOD DONOR FINDER

1. Register Donor
2. View All Donors
3. Search by Blood Group
4. Search by City
5. Exit

Enter your choice: 3

Enter required blood group: A+

===== SEARCH RESULT =====

Name : Arun
 Age : 25
 Gender : Male
 Blood Group: A+
 City : Chennai
 Phone : 9876543210

Name : Divya
 Age : 24
 Gender : Female

Blood Group: A+

City : Panruti

Phone : 9876523456